

Week 12

1. Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$. Find A^{50} .

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 1-\lambda & 0 \\ -1 & -1 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda)^2 \leadsto \lambda_1 = 2, \lambda_2 = 1 \rightarrow A = \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \overset{\text{inverse}}{\begin{bmatrix} -1/2 & -1/2 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}$$

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & -1 & -1 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \leadsto x_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad (\text{check: } Ax_1 = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = 2x_1 \checkmark)$$

$$A - I = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & -1 & 0 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \leadsto x_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad (\text{check } Ax_2 = x_2, Ax_3 = x_3 \checkmark)$$

$$A^{50} = \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2^{50} & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} -1/2 & -1/2 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2^{50} & 2^{50}-1 & 0 \\ 0 & 1 & 0 \\ -2^{50}+1 & -2^{50}+1 & 1 \end{bmatrix}$$

2. Consider the migration matrix

$$A = \begin{bmatrix} 0.6 & 0 & 0.1 \\ 0.2 & 0.8 & 0 \\ 0.2 & 0.2 & 0.9 \end{bmatrix} \leadsto \text{columns add to 1} \Rightarrow A \text{ is a Markov matrix} \Rightarrow \text{has an e-val 1.}$$

for locations 1, 2, and 3. Suppose initially there are 100 residents in location 1, 200 in location 2 and 400 in location 3. Find the population in the three locations after a long time.

Find a steady state vector: $A - I = \begin{bmatrix} -0.4 & 0 & 0.1 \\ 0.2 & -0.2 & 0 \\ 0.2 & 0.2 & -1 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & -1/4 \\ 0 & 0 & 0 \end{bmatrix} \leadsto x = \begin{bmatrix} 1/4 \\ 1/4 \\ 1 \end{bmatrix} \quad (\text{check!})$

↑
adds up to $\frac{3}{2}$

initial condition $u_0 = \begin{bmatrix} 100 \\ 200 \\ 400 \end{bmatrix}$

↑
adds up to 700

→ scale x so that it adds up to 700:

$$\frac{1400}{3} x = \begin{bmatrix} 350/3 \\ 350/3 \\ 1400/3 \end{bmatrix} \begin{matrix} \leftarrow \text{population in loc 1} \\ \leftarrow \text{population in loc 2,} \\ \leftarrow \text{population in loc 3} \end{matrix}$$

after a long time!

3.

Suppose a dynamical system is of the form

$$\begin{bmatrix} x(n+1) \\ y(n+1) \end{bmatrix} = \begin{bmatrix} 1.5 & -0.5 \\ 1.0 & 0 \end{bmatrix} \begin{bmatrix} x(n) \\ y(n) \end{bmatrix}$$

Find solutions to the dynamical system for initial conditions $x_0 = 20, y_0 = 10$.

$$\text{Let } A = \begin{bmatrix} 1.5 & -0.5 \\ 1 & 0 \end{bmatrix} \xrightarrow{\text{diagonalize}} A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\Rightarrow A^n = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/2^n \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 - \frac{1}{2^n} & \frac{1}{2^n} - 1 \\ 2 - \frac{1}{2^{n+1}} & \frac{1}{2^{n+1}} - 1 \end{bmatrix}$$

$$A^n \begin{bmatrix} 20 \\ 10 \end{bmatrix} = \begin{bmatrix} 30 - \frac{10}{2^n} \\ 30 - \frac{10}{2^{n+1}} \end{bmatrix} \rightsquigarrow \begin{cases} x_n = 30 - \frac{10}{2^n} \\ y_n = 30 - \frac{10}{2^{n+1}} \end{cases}$$

4.

Suppose a migration matrix is given by

$$A = \begin{bmatrix} \frac{1}{5} & \frac{1}{2} & \frac{1}{5} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{11}{20} & \frac{1}{4} & \frac{3}{10} \end{bmatrix} \rightsquigarrow \text{columns add to 1} \Rightarrow A \text{ is a Markov matrix} \Rightarrow A \text{ has an e-val 1.}$$

Find the comparison between the population in the three location after a long time.

$$\text{Find a steady state vector: } A - I = \begin{bmatrix} -4/5 & 1/2 & 1/5 \\ 1/4 & -3/4 & 1/2 \\ 11/20 & 1/4 & -7/10 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -16/19 \\ 0 & 1 & -18/19 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow X = \begin{bmatrix} 16/19 \\ 18/19 \\ 1 \end{bmatrix}$$

entries of X add up to $53/19$. In order to find proportions, scale it so that it adds up to 1.

$$\Rightarrow \frac{19}{53} X = \begin{bmatrix} 16/53 \\ 18/53 \\ 19/53 \end{bmatrix} \begin{cases} \leftarrow 16/53 \text{ of population will be in loc 1} \\ \leftarrow 18/53 \text{ of population will be in loc 2} \\ \leftarrow 19/53 \text{ of population will be in loc 3} \end{cases}$$

after a long time

Step-by-step reduction:

or

$$\begin{pmatrix} -\frac{4}{5} & \frac{1}{2} & \frac{1}{5} \\ \frac{1}{4} & -\frac{3}{4} & \frac{1}{2} \\ \frac{11}{20} & \frac{1}{4} & -\frac{7}{10} \end{pmatrix}$$

$$\underline{-\frac{5}{4} R_1 \rightarrow}$$

$$\begin{pmatrix} 1 & -\frac{5}{8} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{3}{4} & \frac{1}{2} \\ \frac{11}{20} & \frac{1}{4} & -\frac{7}{10} \end{pmatrix}$$

$$\underline{R_2 - \frac{1}{4} R_1 \rightarrow}$$

$$\begin{pmatrix} 1 & -\frac{5}{8} & -\frac{1}{4} \\ 0 & -\frac{19}{32} & \frac{9}{16} \\ \frac{11}{20} & \frac{1}{4} & -\frac{7}{10} \end{pmatrix}$$

$$\underline{R_3 - \frac{11}{20} R_1 \rightarrow}$$

$$\begin{pmatrix} 1 & -\frac{5}{8} & -\frac{1}{4} \\ 0 & -\frac{19}{32} & \frac{9}{16} \\ 0 & \frac{19}{32} & -\frac{9}{16} \end{pmatrix}$$

$$\underline{-\frac{32}{19} R_2 \rightarrow}$$

$$\begin{pmatrix} 1 & -\frac{5}{8} & -\frac{1}{4} \\ 0 & 1 & -\frac{18}{19} \\ 0 & \frac{19}{32} & -\frac{9}{16} \end{pmatrix}$$

$$\underline{R_3 - \frac{19}{32} R_2 \rightarrow}$$

$$\begin{pmatrix} 1 & -\frac{5}{8} & -\frac{1}{4} \\ 0 & 1 & -\frac{18}{19} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\underline{R_1 + \frac{5}{8} R_2 \rightarrow}$$

$$\begin{pmatrix} 1 & 0 & -\frac{16}{19} \\ 0 & 1 & -\frac{18}{19} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\underline{R_2 + \frac{5}{16} R_1 \rightarrow}$$

$$\begin{pmatrix} -\frac{4}{5} & \frac{1}{2} & \frac{1}{5} \\ \frac{1}{4} & -\frac{3}{4} & \frac{1}{2} \\ \frac{11}{20} & \frac{1}{4} & -\frac{7}{10} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{4}{5} & \frac{1}{2} & \frac{1}{5} \\ 0 & -\frac{19}{32} & \frac{9}{16} \\ \frac{11}{20} & \frac{1}{4} & -\frac{7}{10} \end{pmatrix}$$

$$\underline{R_3 + \frac{11}{16} R_1 \rightarrow}$$

$$\begin{pmatrix} -\frac{4}{5} & \frac{1}{2} & \frac{1}{5} \\ 0 & -\frac{19}{32} & \frac{9}{16} \\ 0 & \frac{19}{32} & -\frac{9}{16} \end{pmatrix}$$

$$\underline{R_3 + R_2 \rightarrow}$$

$$\begin{pmatrix} -\frac{4}{5} & \frac{1}{2} & \frac{1}{5} \\ 0 & -\frac{19}{32} & \frac{9}{16} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\underline{-\frac{32}{19} R_2 \rightarrow}$$

$$\begin{pmatrix} -\frac{4}{5} & \frac{1}{2} & \frac{1}{5} \\ 0 & 1 & -\frac{18}{19} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\underline{R_1 - \frac{1}{2} R_2 \rightarrow}$$

$$\begin{pmatrix} -\frac{4}{5} & 0 & \frac{64}{95} \\ 0 & 1 & -\frac{18}{19} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\underline{-\frac{5}{4} R_1 \rightarrow}$$

$$\begin{pmatrix} 1 & 0 & -\frac{16}{19} \\ 0 & 1 & -\frac{18}{19} \\ 0 & 0 & 0 \end{pmatrix}$$